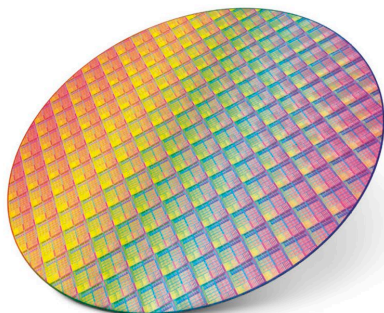


Bronze

The metal tin was one of the earliest metals to be purified. For thousands of years, people have mixed tin with copper to create an alloy called bronze, which is tougher and more useful than either of the original metals. The first widespread human civilizations used bronze to make tools, statues, and weapons.

6th century BCE bronze
statue of two oxen



Semiconducting silicon wafer

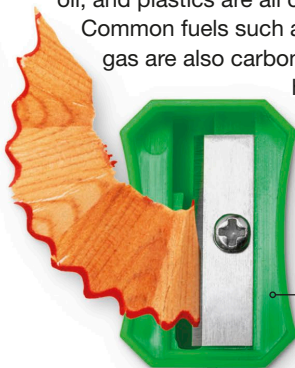
Semiconductors

The impure forms of germanium and silicon are used as semiconductors – substances that conduct electricity better than insulators, but not as well as conductors. When pure, these two semi-metals block electricity, but when tiny amounts of other elements are added to them, they can transmit it. These semiconductors are used in electronics.

Carbon chemistry

One carbon atom can bond to anything from one to four other atoms, and as a result, carbon forms a huge range of compounds. Wood, coal, oil, and plastics are all carbon compounds.

Common fuels such as petrol and natural gas are also carbon-based. They release heat when burned, and transform into water and carbon dioxide, as well as carbon monoxide, which is a toxic gas.



The body of this sharpener is made of plastic and the steel blade is toughened with carbon grains.

This wood is made of cellulose, a carbon compound.

The pencil lead is graphite, a form of pure carbon.

Pencil with
plastic sharpener